SF2940 Probability: Exam

Fall 2022

Date: 21. October 2022, 08:00-12:00.
Examiner: Kevin Schnelli, SCI MAT. Phone: 72 02
Aids: One page hand written notes. No other aids are allowed. Formulary with distributions is included after the problem statement.

General instructions: Your solutions and answers should be legible, easy to follow and not lack in explanations. For full points a correct and well-presented reasoning is required.

Exam design: This exam consists of five independent problems. Each problem gives at most 5 points.

Other: Solutions will be posted on Canvas within a week after the exam.

Problem 1. Let X be a random variable such that $X|A = a \in U(0,a)$ where the random variable A has probability density function

$$f_A(a) = \begin{cases} 6a(1-a)\,, & \text{if } 0 < a < 1\,, \\ 0\,, & \text{else}\,. \end{cases}$$

a.) Determine $\mathbb{E}[X|A]$ and $\mathbb{E}[X]$. ..... (2points)

b.) Determine the probability density function $f_X(x)$ of the random variable X. Make sure to be precise in your reasoning.
..... (3points)

Problem 2. Let $(X_i)_{i\geq 1}$ be a sequence of i.i.d. random variables with $\mathbb{E}[X_1] = 0$ and $\mathrm{Var}(X_1) = 2$, and let $N \in \mathrm{Po}(\lambda)$, $\lambda > 0$, be an independent Poisson random variable.

a.) Show that

$$\frac{N}{\lambda} \xrightarrow{P} 1\,, \qquad \text{as } \lambda \to \infty.$$

..... (1point)

b.) Determine the limit in distribution of

$$\frac{N-\lambda}{\sqrt{\lambda}}\,,$$

as $\lambda \to \infty$..... (2points)

c.) Set now $S_n := \sum_{i=1}^n X_i$, $n \geq 1$, and $S_0 := 0$. Determine the limit in distribution of

$$\frac{S_N}{\sqrt{\lambda}}\,,$$

as $\lambda \to \infty$..... (2points)

Hint: Characteristic function.

Problem 3. Let $\mathbf{W} = \{W(t) : t \geq 0\}$ be a Wiener process.

a.) Compute the conditional expectation $\mathbb{E}\big[(W(1))^2 + W(2)|W(1)\big]$ ..... (1point)

b.) Determine the distribution of the random vector $(W(1), W(2), W(3) - W(1))'$. ..... (2points)

c.) We introduce another stochastic process $\mathbf{X} = \{X(t) : t \geq 0\}$ by setting

$$X(t) := \mathrm{e}^{W(t) - \frac{t}{2}}\,, \quad t \geq 0.$$

Compute $\mathbb{E}[X(t)|W(s)]$, for $t \geq s \geq 0$.
Hint: For $Y \in \mathcal{N}(0, \sigma^2)$, its moment generating function is given by $\psi_Y(r) := \mathbb{E}[\mathrm{e}^{rY}] = \mathrm{e}^{\sigma^2 r^2/2}$, $r \in \mathbb{R}$. ..... (2points)

Problem 4. Let $(X_i)_{i\geq 1}$ be a sequence of i.i.d. random variables with $\mathbb{E}[X_1] = \mu$ and $\mathrm{Var}(X_1) = \sigma^2$, for given $\mu \in \mathbb{R}$, $\sigma^2 > 0$. Set now

$$S_n := \sum_{i=1}^{n} X_i\,, \qquad n \geq 1\,,$$

and $S_0 := 0$.

Determine the limit in distribution of the sequence

$$n^{5/2}\,\frac{\exp\left(\frac{S_n}{n}\right) - \exp\left(\mu\right)}{n\sum_{i=1}^{n}(X_i)^2 - (S_n)^2}\,,$$

as $n \to \infty$. Here, exp denotes the exponential function. Justify each step of your reasoning. ..... (5points)

Problem 5. Let $\mathbf{X} = \{X(t) : t \in [0, \infty)\}$ be a stochastic process with the following three properties:

1.) $X(0) = 0$ almost surely.

2.) The increments $\{X(t_k) - X(t_{k-1})\}$, $1 \leq k \leq n$, are independent random variables, for all $0 \leq t_0 \leq t_1 \leq t_2 \leq \cdots \leq t_n$, and all $n \in \mathbb{N}$.

3.) For $0 \leq s < t$,

$$(X(t) - X(s)) \in \mathrm{Po}(t - s)\,,$$

that is $(X(t) - X(s))$ is Poisson distributed with parameter $t - s$.

Recall that if $Y \in \mathrm{Po}(\lambda)$, $\lambda > 0$, then $\mathrm{Var}(Y) = \lambda$.

a.) Compute $\mathbb{E}[X(t)|X(s)]$, for $t \geq s \geq 0$. . . . . (2points)

b.) Compute the autocorrelation function $R_{\mathbf{X}}(t, s)$, for all $t, s \geq 0$.
Hint: Assume first that $t > s$, then consider $t \leq s$. . . . . (2points)

c.) It follows immediately from the properties 1.)–3.) above that $X(t)$ takes values in the non-negative integers. Compute thus the probability generating function $g_{X(t)}(r)$ of $X(t)$, for all $t \geq 0$, $r \in \mathbb{R}$. . . . . (1point)

Good luck!

SF2940: Distributions and their characteristic functions

Distribution	Notation	Probability function	$\varphi_X(t)$
One point	$\delta(a)$, $a \in \mathbb{R}$	$p(a) = 1$	$\mathrm{e}^{\mathrm{i}ta}$
Bernoulli	$\mathrm{Be}(p)$; $0 \le p \le 1$	$p(0) = q$, $p(1) = p$; $q = 1 - p$	$q + p\mathrm{e}^{\mathrm{i}t}$
Binomial	$\mathrm{Bin}(n, p)$, $n = 1, 2, \ldots$; $0 \le p \le 1$	$p(k) = \binom{n}{k} p^k q^{n-k}$, $k = 0, 1, 2, \ldots, n$; $q = 1 - p$	$(q + p\mathrm{e}^{\mathrm{i}t})^n$
Geometric	$\mathrm{Ge}(p)$, $0 \le p \le 1$	$p(k) = pq^k$, $k = 0, 1, 2, \ldots, n$; $q = 1 - p$	$\frac{p}{1-q\mathrm{e}^{\mathrm{i}t}}$
Poisson	$\mathrm{Po}(\lambda)$, $\lambda > 0$	$p(k) = \mathrm{e}^{-\lambda}\frac{\lambda^k}{k!}$, $k = 0, 1, 2, \ldots$	$\mathrm{e}^{\lambda(\mathrm{e}^{\mathrm{i}t}-1)}$

Table 1. Discrete distributions

Distribution	Notation	Density	$\varphi_X(t)$
Uniform	$\mathrm{U}(a, b)$, $a < b$	$f(x) = \frac{1}{b-a}$, $a < x < b$	$\frac{\mathrm{e}^{\mathrm{i}tb}-\mathrm{e}^{\mathrm{i}ta}}{\mathrm{i}t(b-a)}$
Exponential	$\mathrm{Exp}(\lambda)$, $\lambda > 0$	$f(x) = \frac{1}{\lambda}\mathrm{e}^{-x/\lambda}$, $x > 0$	$\frac{1}{1-\lambda\mathrm{i}t}$
Laplace	$\mathrm{L}(\lambda)$, $\lambda > 0$	$f(x) = \frac{1}{2\lambda}\mathrm{e}^{-\lvert x\rvert/\lambda}$, $x \in \mathbb{R}$	$\frac{1}{1+\lambda^2t^2}$
Beta	$\beta(r, s)$, $r, s > 0$	$f(x) = \frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)}x^{r-1}(1-x)^{s-1}$, $0 < x < 1$	$*$
Normal	$\mathrm{N}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\mathrm{e}^{-\frac{1}{2}(x-\mu)^2/\sigma^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{\mathrm{i}\mu t-\frac{1}{2}\sigma^2t^2}$
Standard normal	$\mathrm{N}(0, 1)$	$f(x) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-\frac{1}{2}x^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{-\frac{1}{2}t^2}$
Log-normal	$\mathrm{LN}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$	$f(x) = \frac{1}{x\sqrt{2\pi\sigma^2}}\mathrm{e}^{-\frac{1}{2}(\log x-\mu)^2/\sigma^2}$, $x > 0$	$*$
Cauchy	$\mathrm{C}(m, \eta)$, $\eta > 0$, $m \in \mathbb{R}$	$f(x) = \frac{1}{\pi}\frac{\eta}{(x-m)^2+\eta^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{\mathrm{i}mt-\eta\lvert t\rvert}$

Table 2. Continuous distributions

$*$ indicates either a too complicated expression or no explicit expression exists.
Date September 22, 2021